



Hybrid Expert Knowledge and Self-Supervised Learning for Diagnostic Modeling of Adductor Spasmodic and Primary Myotonic Dysphonia

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Abstract

Dysphonia encompasses a broad spectrum of vocal disorders with diverse etiologies, among which adductor spasmodic dysphonia (ADSD) and primary muscle tension dysphonia (pMTD) are particularly challenging to diagnose. Currently, the primary diagnostic method relies on subjective auditory perception by highly experienced clinicians. To alleviate the scarcity of diagnostic resources, this study develops a deep learning-based approach for automatically diagnosing ADSD and pMTD using patients' speech data. Our contributions are: (1) designing a convolutional neural network (CNN)-based diagnostic model that leverages handcrafted features derived from expert knowledge and (2) incorporating self-supervised learning (SSL) to extract more discriminative representations as input from raw waveforms adaptively. This marks the first application of deep learning techniques to ADSD and pMTD diagnostic modeling, achieving a classification accuracy of 83.3% on our newly constructed dataset.

Index Terms: speech disorder classification, self-supervised learning, diagnostic speech processing

1. Introduction

As is well known, accurate disease diagnosis is crucial for subsequent treatment plans [1]. Misdiagnosis can lead to inappropriate treatments, potentially worsening the condition. However, adductor spasmodic dysphonia (ADSD) and primary muscle tension dysphonia (pMTD) are often confused and misdiagnosed [2, 3, 4]. This is because the clinical diagnosis of these two conditions primarily relies on subjective auditory perception [5], yet their acoustic features are very similar, making it difficult for many non-specialist specialists to distinguish between them [6]. In this context, there is a growing demand for automated diagnostic methods based on acoustic signals.

In recent years, research on the automatic differential diagnosis of ADSD and pMTD has gradually evolved from traditional acoustic parameter analysis to multidimensional feature fusion and algorithm-driven approaches. Early studies based on long-term average spectra (LTAS) spectral moment parameters differentiated ADSD from pMTD, revealing the dispersed low-frequency energy distribution characteristic of ADSD [7], which belongs to steady-state speech segment analysis. Roy [8] further found significant dynamic differences in fundamental frequency (F0) and glottal closure features between ADSD patients during reading and conversation tasks, while pMTD patients exhibited relatively stable features. Li [9] integrated a creak recognition model that incorporates prosodic position,

Table 1: A comparison of previous and our datasets.

Models	Content	pMTDs	ADSDs	Durations
LTAS [7]	sustained vowels	59	41	30min
CSID [8]	/a/ and one sentence	45	36	50min
Creak% [9]	six sentences	16	16	50min
Ours	/a/ /i/ and six sentences	40	40	200min

pitch, and pitch range. An open source creak detector was employed to compute the percentage of creak (Creak%) in each of the six recorded sentences of the speakers, validating the complementary role of short-term perturbations and long-term prosodic features. A recent study found that the Cepstral Spectral Index of Dysphonia (CSID) [8] outperformed Creak% [9] and manual auditory assessment in distinguishing ADSD, with superior diagnostic accuracy, particularly on the normalized voiceless sentence [10]. However, existing work focuses on exploring the interpretability of limited features and verifying their effectiveness with shallow classifiers, such as SVM and random forests [7, 8, 9]. There remains significant room for improvement in the characterization of disease heterogeneity.

Recently, deep learning techniques [11] have significantly improved various tasks driven by large-scale datasets. However, due to the relative rarity of ADSD patients or challenges in diagnosis [12], existing datasets have relatively limited patients and audio contents, as shown in Table 1. LTAS includes sentences with all vowels, CSID primarily focuses on sustained vowels /a/ and read sentences, while Creak% covers the first six sentences of the Rainbow Passage. This study is the first to construct a Mandarin speech dataset for ADSD and pMTD patients, combining vowels and six sentences, totaling 200 minutes in duration, with data balance between the pMTD and ADSD groups. Moreover, the case annotations were completed by senior experts with certification as licensed speech-language pathologists (CCC-SLP) in the United States.

To address the challenge of differential diagnosis between these two disorders, we first designed a method that employs a CNN framework with multidimensional features from openS-MILE [13]. Then, we integrated a CNN-based diagnosis model with a self-supervised pre-trained feature extractor (such as WavLM [14]). Additionally, considering that pre-trained models derived from different speech tasks may not be well-suited for the current specific task, we incorporated adapter modules into the self-supervised model to better adjust it for downstream tasks. Extensive experiments and comprehensive comparisons on a custom-built dataset validate the effectiveness of the CNN model and SSL features in diagnosing pMTD and ADSD. The contributions are summarized as follows:

- designing a CNN-based diagnostic model that leverages mul-

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tidimensional features derived from acoustic knowledge;

- incorporating an SSL-based feature extractor with adapters into the aforementioned diagnostic model to handle limited training samples;
- verifying the effectiveness of the CNN-SSL diagnostic model through a series of experiments on a real-world pMTD and ADSD corpus.

2. Related Work

The differential diagnosis of ADSD and pMTD remains challenging. Early studies [6] pointed out that acoustic and aerodynamic parameters can be helpful in distinguishing ADSD, MTD, and vocal cord tremor. Rees [12] demonstrated that experienced speech pathologists can identify voice disruptions, irregular vertical striations, and retained resonance peaks through spectral analysis, validating its potential for diagnosing dynamic phonatory patterns. Yan [15] proposed a spatiotemporal analysis method using high-speed videoendoscopy, quantifying the asymmetry and inhomogeneity of vocal cord vibration patterns via Nyquist diagrams, along with glottal area waveform analysis, successfully distinguishing symmetric vibrations in MTD from intermittent spasms in ADSD, providing a new paradigm for the objective assessment of dynamic vibrational patterns. Carroll [16] further explored aerodynamics and revealed that ADSD patients show higher glottal peak pressure and final airflow in /p/ syllable tasks, while pMTD patients, especially those with pMTD, have significantly reduced final airflow, providing a quantitative basis for the aerodynamic features of these two diseases. Chen [17] used high-speed videoendoscopy (HSV) combined with digital kymography analysis to reveal unique oscillatory breaks in the onset phase of vocal fold vibration in ADSD patients, while MTD patients mainly exhibited excessive vocal fold tension with sustained supraglottic compression, providing a new method for visualizing and distinguishing vibratory patterns.

Other research has focused on exploring either ADSD or pMTD individually. By evaluating the IINVo perceptual scale and AMPEx acoustic analysis, finding their complementary roles helped identify ADSD-specific voice disruptions and fundamental frequency irregularities [18]. A work used spatiotemporal processing to quantify vocal fold vibration irregularities in ADSD and distinguish them from neurological disorders [15]. Latest work finding that Maximum Phonation Time (MPT) effectively distinguishes SD subtypes, with ADSD patients having significantly longer MPT than controls [19]. For MTD detection, Past work shows that MTD patients exhibited posterior glottal gaps, and MPT was reduced compared to other acoustic and analysis methods [3]. Another study found significant correlations between laryngeal features and abnormal subglottic pressure peaks, showing statistical significance [20]. One of the latest efforts used random forest models and multivariate statistical analysis to reveal that the 15 acoustic and aerodynamic parameters recommended by the American Speech-Language-Hearing Association, overlap significantly when distinguishing pMTD from other voice disorders, with a classification accuracy of only 65%. This indicates limitations in the traditional acoustic parameters for characterizing the heterogeneity and specificity of pMTD, suggesting the need for combining physiological dynamics to improve diagnostic accuracy [21].

The automatic diagnosis of pathological voice disorders is receiving increasing attention, particularly with the application of deep learning algorithms in speech disorder classifica-

tion. A review pointed out that the trend of combining self-supervised learning with acoustic features offers new perspectives for speech pathology applications [22]. A study demonstrated the advantages of the pre-trained Wav2Vec 2.0 model for cross-database speech disorder detection[23]. Another study proposed a hybrid model combining Mel frequency cepstral coefficient and other acoustic features to enhance the robustness of detection [24]. And improved classification accuracy by combining CNN with transfer learning [25]. One of the latest efforts presented a SSL approach that significantly improved speech disorder detection efficiency and scalability with a small amount of labeled data [26].

3. Dataset

The audio data in this study were collected from 80 participants at the Voice Evaluation Room of the Eye and ENT Hospital of Fudan University. The dataset consists of 40 ADSD patients and 40 pMTD patients, with detailed demographic information provided in Table 2. It is divided into three subsets: training, validation, and evaluation, with statistical analysis conducted on the number of sentences, speech duration, and gender distribution of participants. Additionally, the dataset includes the total number of speech samples and gender distribution of the participants. The study was approved by the Ethical Review Committee of the Eye and ENT Hospital of Fudan University (IRB2021060).

Table 2: *Demographic information of subjects*

Dataset	Train	Val	Eval	Total
Duration (min)	140	40	20	200
Utterance	1,008	288	144	1,440
Participant	56	16	8	80
-Male	21	6	3	31
-Female	34	10	5	49

4. Method

4.1. Framework overview

As shown in Figure 1, the proposed framework integrates expert knowledge-driven feature extraction with SSL to improve the differential diagnosis of ADSD and pMTD. The architecture consists of two parallel pathways: CNN-EK, which captures handcrafted features derived from domain expertise, and CNN-SSL, which leverages self-supervised representations to adaptively extract salient speech patterns.

In the CNN-EK branch, openSMILE extracts a comprehensive set of handcrafted acoustic features, which are subsequently processed through a series of one-dimensional convolutional layers to capture temporal dependencies. Meanwhile, the CNN-SSL branch directly operates on raw speech waveforms, utilizing a pre-trained WavLM model with adapter modules to refine feature representations for the specific classification task. The extracted representations are further processed through additional one-dimensional convolutional layers to enhance their discriminative capability. By combining expert knowledge with data-driven representation learning, this dual-branch framework effectively balances interpretability and adaptability. The final classification is achieved through a fully connected layer with a Sigmoid activation function, ensuring robust prediction results.

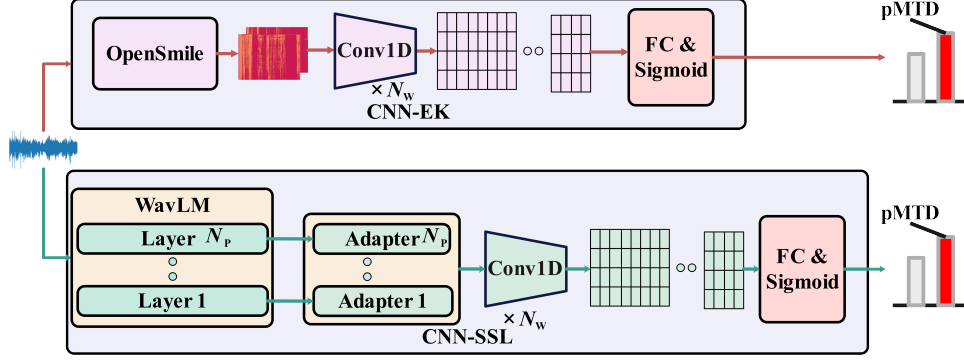


Figure 1: Framework of CNN-EK and CNN-SSL models

4.2. Model Architecture

4.2.1. CNN-EK Module

In the CNN-EK branch, a deep neural network is employed to process multidimensional handcrafted features derived from expert knowledge. First, the openSMILE toolkit is used to extract various acoustic features from the audio data, including Mel frequency cepstral coefficients (MFCCs), root mean square energy, zero-crossing rate, and fundamental frequency [27]. By extracting high-dimensional features, the model can effectively leverage the rich information embedded in these handcrafted representations. The CNN-EK extracts features as follows:

(1) Short-Term Low-Level Descriptor (LLD) Extraction: Frame-based acoustic features are computed, including:

- Mel frequency spectral coefficients:

$$c_n = \sum_{k=1}^K \log E_k \cdot \cos \left(n \left(k - \frac{1}{2} \right) \frac{\pi}{K} \right), \quad n = 1, \dots, 15 \quad (1)$$

where E_k denotes Mel filterbank energy.

- Log Mel Spectrum:

$$S_{\text{mel}}(m) = \ln \left(\sum_{k=0}^{N-1} |X(k)|^2 H_m(k) \right) \quad (2)$$

with triangular Mel-scale filterbank $H_m(k)$ to approximate human auditory perception.

- Spectral Features: Linear Prediction Spectral coefficients and root mean square energy.

(2) Statistical Functional Application: Window-level aggregation via 3 base functionals (μ, σ, Δ) and 19 extended functionals (e.g., quartiles):

$$Q(p) = x_{\lceil pN \rceil}, \quad p \in \{0.25, 0.75\} \quad (3)$$

The total dimensionality of the feature is derived as:

$$N_{\text{Total}} = 1582 = \sum_{g=1}^G \left(N_{\text{LLD}}^{(g)} \times N_{\text{func}}^{(g)} \right) \quad (4)$$

(3) Pitch-Centric Characterization: Supra-segmental dynamics are modeled through:

- Fundamental Frequency: Detected by autocorrelation:

$$F0 = \underset{\tau}{\operatorname{argmax}} \sum_{t=0}^{T-1} s(t)s(t+\tau) \quad (5)$$

The extracted features (1582-D vector from Equation 4) serve as input feature vectors for the deep neural network, enabling hierarchical encoding of both frame-level acoustic patterns (e.g., MFCCs in Equation 1) and utterance-level prosodic trends (e.g., F0 contours in Equation 5). The final layer consists of a fully connected layer followed by a Sigmoid activation function to produce classification results.

4.2.2. CNN-SSL Module

In the CNN-SSL system, a self-supervised model based on WavLM is adopted [28]. The model architecture consists of a series of convolutional layers, feature projection layers, and encoder layers designed for processing and classifying speech data. It is initialized with a pre-trained WavLM feature encoder, which extracts high-dimensional features directly from raw speech waveforms. The feature extraction module comprises a sequence of convolutional layers, each employing a GELU activation function, along with normalization and dropout techniques to refine layer outputs. The model includes seven convolutional layers, each designed to reduce dimensionality while preserving key speech information.

Following feature extraction, the model employs an encoder with self-attention mechanisms and feedforward layers, combined with position-based convolutional embeddings, to maintain the sequential characteristics of speech data. Additionally, adapter modules are incorporated to fine-tune each layer. These adapters modify the learned features from the WavLM model to better suit the specific classification task. The adapter operates hierarchically, where higher layers are fine-tuned using progressively narrower task-specific adapters, ensuring optimal feature customization for downstream tasks.

After adaptation, the feature maps are further refined using a one-dimensional convolutional neural network (CNN), enabling the model to capture temporal dependencies and extract discriminative patterns required for classification. This approach effectively harnesses the power of SSL while minimizing the dependence on large labeled datasets, thereby enhancing model performance in low-resource scenarios.

5. Experiments

5.1. Experimental Setup

5.1.1. Baselines

To compare with previous work, we select the following models as baselines:

- LTAS [7] Using the spectral matrix as feature input and com-

pleting classification with a logistic regression classifier.

- CSID [8] Using the spectral matrix as a feature input and completing the classification with a logistic regression classifier.
- Creak% [9] Using vocal fry as the sole feature, combined with a threshold classifier to complete the classification task.

5.1.2. Evaluation Metrics and Strategy

We evaluate the performance of the model using three widely recognized metrics: accuracy, specificity, and sensitivity. These metrics offer a thorough evaluation of the model's performance, with sensitivity and specificity highlighting how effectively the model identifies positive and negative cases in binary classification tasks. The dataset is divided into training, validation, and test sets with a 0.7:0.2:0.1 ratio. To ensure reliable results, the mean and standard deviation of test performance are reported in multiple runs.

5.2. Experimental Results

In our proposed CNN-SSL model, we conducted an ablation study on four mainstream speech pre-trained models: Whisper [29], SpeechTokenizer [30], Wav2Vec2 [31], and WavLM [14], as shown in Table 3. WavLM achieved the best performance across all three metrics, with an accuracy of 0.833 and a sensitivity of 0.875, representing a relative improvement of 20.9% and 20.7%, respectively, compared to the baseline model Whisper. Notably, while WavLM has a larger parameter size, the smaller Wav2Vec2 model still achieved an accuracy of 0.779, outperforming SpeechTokenizer and Whisper by 7.7% and 13.1%, respectively. Therefore, WavLM is more suitable for the classification of ADSD and pMTD. Consequently, we selected WavLM as the pre-trained model for the CNN-SSL framework in subsequent experiments.

Table 3: Performance comparison of Whisper, SpeechTokenizer, Wav2vec2, and WavLM.

Models	Accuracy	Specificity	Sensitivity
Whisper [29]	0.689	0.650	0.725
SpeechTokenizer [30]	0.723	0.700	0.745
Wav2vec2 [31]	0.779	0.758	0.800
WavLM [14]	0.833	0.791	0.875

To investigate the impact of different vowels on model performance, we conducted an additional ablation study. As shown in Table 4, the CNN-SSL model based on WavLM outperformed CNN-EK in overall classification accuracy. However, further analysis revealed performance differences across vowels. Specifically, CNN-EK achieves better results on the vowel /i/, while CNN-SSL slightly outperforms on the vowel /a/. In addition, both CNN-EK and CNN-SSL outperform on the vowel /a/ than on the vowel /i/, and the best results were obtained when both vowels were combined.

We conducted a comprehensive performance evaluation to validate the classification effectiveness of the proposed deep learning models. As shown in Table 5, both CNN-EK and CNN-SSL achieved substantial improvements over traditional methods. CNN-SSL demonstrated the highest classification performance, achieving an accuracy of 0.833 and a sensitivity of 0.875. CNN-EK also performed strongly, with an accuracy of 0.818 and a sensitivity of 0.862. In contrast, traditional ap-

Table 4: Comparison of the classification performance of two vowel pairs models

Model	/a/	/i/	Accuracy	Specificity	Sensitivity
CNN-EK	✓		0.779	0.738	0.820
		✓	0.742	0.688	0.795
	✓	✓	0.791	0.746	0.835
CNN-SSL	✓		0.782	0.737	0.828
		✓	0.731	0.680	0.782
	✓	✓	0.807	0.764	0.850

Table 5: The comparison of prediction performance. The best results are bold.

Methods	Accuracy	Specificity	Sensitivity
LTAS [7]	0.583	0.500	0.600
CSID [8]	0.663	0.660	0.680
Creak% [9]	0.663	0.670	0.640
CNN-EK	0.818	0.774	0.862
CNN-SSL	0.833	0.791	0.875

proaches such as LTAS, CSID, and Creak% exhibited lower accuracy, ranging from 0.583 for LTAS to 0.663 for CSID and Creak%, and lower sensitivity, with CSID reaching 0.68 and Creak% at 0.64.

Beyond overall classification performance, the specificity results further emphasize the advantages of CNN-based models, particularly in distinguishing pMTD cases. pMTD has been especially challenging to diagnose due to its greater variability in acoustic presentation, often leading to poorer classification performance in previous studies. CNN-SSL achieved a specificity of 0.791, while CNN-EK reached 0.774, both substantially outperforming CSID at 0.66 and Creak% at 0.67. The improved specificity of CNN-based models indicates a stronger ability to accurately identify pMTD cases, reducing misclassification rates and enhancing diagnostic reliability. These findings demonstrate that deep learning models, particularly CNN-SSL, provide a more robust and precise approach for the differential diagnosis of ADSD and pMTD.

6. Conclusion

This study developed a deep learning-based method for diagnosing ADSD and pMTD from speech data, addressing the limitations of traditional methods that use a limited set of features and shallow classifiers. Specifically, both CNN-EK and CNN-SSL showed excellent performance on a real-world dataset. Ablation experiments showed that the CNN-SSL model outperformed CNN-EK, with adapter modules further improving the performance. Finally, the five-fold cross-validation confirmed that the model outperforms traditional methods in diagnostic accuracy, specificity, and sensitivity, demonstrating its potential to enhance dysphonia diagnosis.

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8. References

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